

Sudarshan Neopane

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RESEARCH INTERESTS

- **Computational Astrophysics:** Core-collapse supernovae, remnants, and compact objects
- **Fluid Dynamics:** Radiation–magnetohydrodynamics, shock interaction, instabilities
- **High-Performance Computing:** GPU/exascale simulations with Flash-X

EDUCATION

- **University of Tennessee Knoxville** Knoxville, TN
Ph.D. in Physics; minor in Computational Sciences Aug 2021 - July 2026 (Expected)
- **University of Massachusetts Dartmouth** North Dartmouth, MA
M.S. in Physics Sep 2019 - Aug 2021
Thesis: *Near-MCh type Ia supernovae from the double-degenerate channel of merging white dwarfs*
- **St. Xavier's College (Affiliated to Tribhuvan University)** Kathmandu, Nepal
B.Sc. in Physics Oct 2014 - Sep 2018
Thesis: *First-Principles Study of van der Waals Interactions between Halogen Molecules (Cl₂ and I₂)*

EMPLOYMENT/ TEACHING AND RESEARCH EXPERIENCE

- **University of Tennessee Knoxville**
Research Assistant (*Supervisor: William R Hix*) Aug 2022 - current
Teaching Assistant (*PHY 231: Electricity and Magnetism*) Aug 2021 - July 2022
- **Los Alamos National Laboratory**
Graduate Researcher (*Division: CCS-7*) June 2024 - Aug 2024
- **University of Massachusetts Dartmouth**
Research Assistant (*Supervisor: Robert Fisher*) Jan 2021 - July 2021
Head Teaching Assistant (*PHY 113: Mechanics*) Sep 2020 - Dec 2020
Instructor (*PHY 114: Electricity and Magnetism*) Jul 2020 - Aug 2020
Teaching Assistant (*PHY 113: Mechanics*) Sep 2019 - May 2020

HONORS AND AWARDS

- ALCF Director's Discretionary Award: 20,000 node hours allocation on Aurora Oct 2025 - April 2026
- Robert Birkhoff Physics Fellowship, UT Knoxville Aug 2021 - July 2023
- Academic Excellence Award, St. Xavier's College, Kathmandu, Nepal Sep 2018
- Golden Jubilee Scholarship, Embassy of India, Kathmandu, Nepal Mar 2015 - Feb 2018

SKILLS SUMMARY

- **Languages:** Fortran, C++, Python, Matlab, Bash
- **Simulation Tools:** Flash-X, Flash4.3, Torch, XNet, Quantum ESPRESSO
- **Visualization:** yt, Visit, Blender, Paraview, Xcrysden
- **HPC Skills:** MPI, OpenMP, Cuda, GDB, DDT
- **HPC Systems:** Aurora (ALCF), Frontier(OLCF), Stampede2 (TACC), Chicoma (ASC)

PRESENTATIONS AND WORKSHOPS

Posters and Talks:

- Iposter for 245th AAS meeting, 13 January, 2025
Title-**3D evolution of Core-Collapse Supernovae into the circumstellar medium**
- Poster for SC24, 19 November, 2024
Title-**Computational Radiation Hydrodynamics with FleCSI**
- Iposter for 243rd AAS meeting, 9 January, 2024
Title-**Long-time evolution of Core-Collapse supernovae**
- Talk for 237th AAS meeting, 13 January, 2021
Title-**Near-Chandrasekhar Mass Type Ia Supernovae from the Double-Degenerate Channel**

Workshops:

- Argonne Training Program on Extreme-Scale Computing (ATPESC), 27 July - 8 August 2025 (St. Charles, US)
- N3AS Summer School, 11-20 July 2025 (Santa Cruz, US)
- International HPC Summer School, 9-14 July 2023 (Atlanta, US)
- XSEDE HPC Workshop: Summer Boot Camp, 2-5 June 2020 (Online)

PUBLICATIONS

In-preparation:

- **Three-dimensional simulation of a core-collapse supernova extending years post-bounce with a realistic nuclear reaction network**
S. Neopane, M. A. Sandoval, W. R. Hix, J. A. Harris, O. E. B. Messer, E. J. Lentz ApJ (In prep.), 2026
Contribution: Developed the code, performed the simulations and analyzed the results
- **Simulating the Effects of Realistic Circumstellar Environments on Stripped Envelope CoreCollapse SN**
S. DeSoto, S. Neopane, W. R. Hix, M. A. Sandoval, O. B. Messer, J. A. Harris, E. Laplace, J. L. Hoffman ApJ (In prep.), 2026
Contribution: Developed the simulation infrastructure, provided regular fixes to the numerical algorithms

Refereed:

- HARD: A Performance Portable Radiation Hydrodynamics Code based on FleCSI Framework**
 J. Loiseau, H. Lim, A. López, M. Baghirzade, S. S. Khan, Y. Kim, S. Neopane, A. Strack, F. Taiyebah, B. K. Bergen SoftwareX, 2025
 (Published in [SoftwareX](#))
 DOI: <https://doi.org/10.1016/j.softx.2025.102441>
Contribution: Added flux limited diffusion approximation for radiation transport; performed scaling studies
- Hydrodynamical simulations favour a pure deflagration origin of the near-chandrasekhar mass supernova remnant 3C 397** MNRAS, 2024
 V. Mehta, J. Sullivan, R. Fisher, Y. Ohshiro, H. Yamaguchi, K. Bhargava, S. Neopane
 (Published in [The Monthly Notices of the Royal Astronomical Society](#))
 DOI: <https://doi.org/10.1093/mnras/stae1559>
Contribution: Ran the initial WD simulations; developed the post-processing pipeline
- Near-Chandrasekhar-Mass Type Ia Supernovae from the Double-Degenerate Channel**
 S. Neopane, K. Bhargava, R. Fisher, M. Ferrari, S. Yoshida, S. Toonen, E. Bravo ApJ, 2022
 (Published in [The Astrophysical Journal](#))
 DOI: <https://doi.org/10.3847/1538-4357/ac3b52>
Contribution: Performed theoretical calculations and numerical simulations; developed the post-processing pipeline to obtain synthetic spectra from FLASH simulation data
- First-Principles study of Van Der Waals Interactions between Halogen Molecules (Cl₂ and I₂)**
 S. Neopane, N. Pantha (Published in the [Journal of Nepal Physical Society](#)) J. Nepal Phys. Soc., 2020
 DOI: <https://doi.org/10.3126/jnphysoc.v5i1.26878>
Contribution: Performed density functional theory based simulations and compared with experiments

REFERENCES

- Dr. William R. Hix** raph@utk.edu
 Professor, UTK; PhD advisor
- Dr. Bronson Messer** bronson@ornl.gov
 Joint Faculty Professor, UTK; PhD co-advisor
- Dr. Anthony Mezzacappa** mezz@utk.edu
 Professor, UTK; Dissertation committee member